Lecture 2 - Frameworks and Methods

COMP1787 Requirements Management

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The need for a framework

To develop **effective** systems in the most **efficient** way possible.

The traditional systems development life cycle (with which you are all familiar) was a first attempt to provide a controlled environment for systems development.

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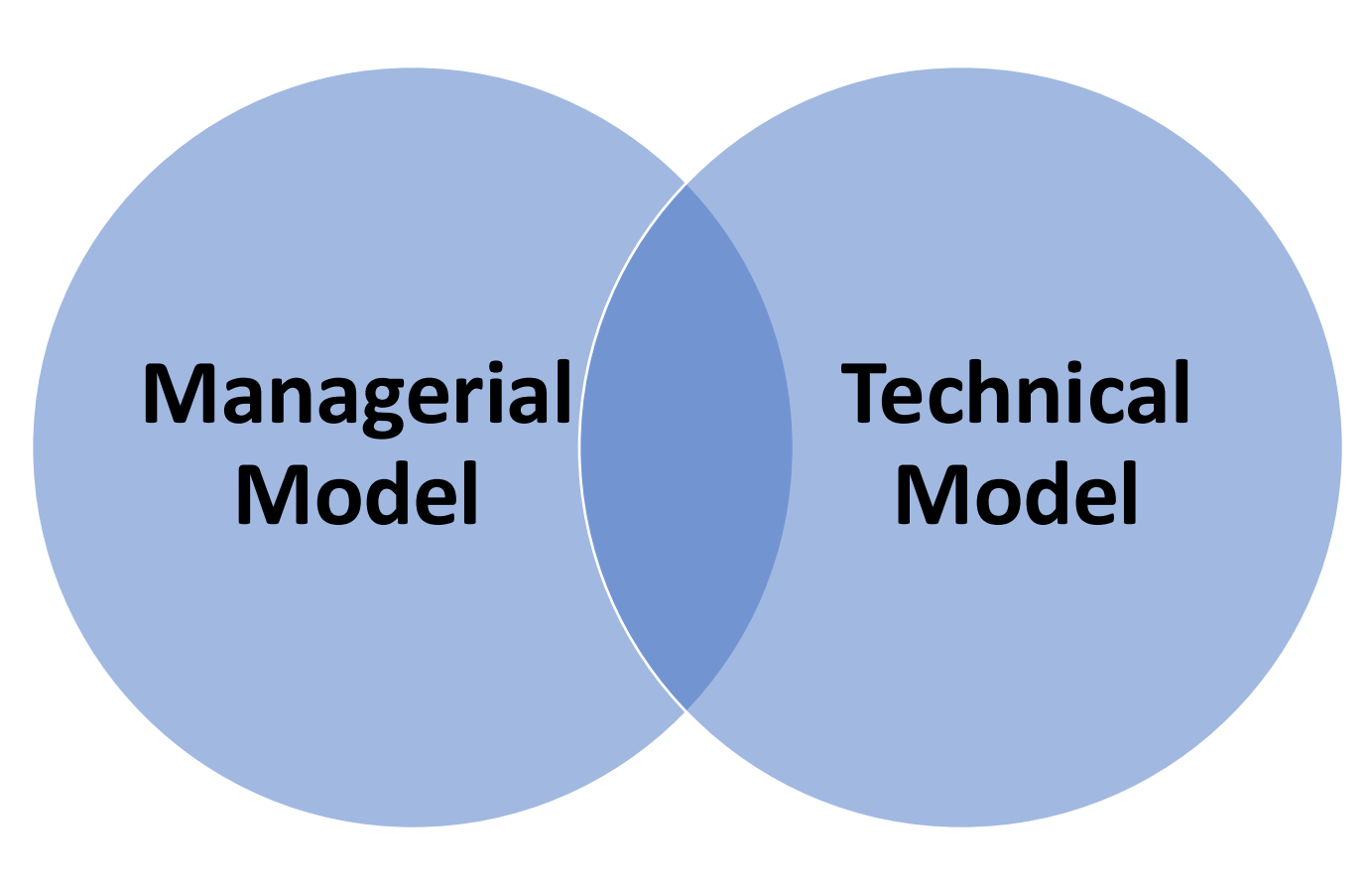
What we need from a systems development methodology

- Generic development framework



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Technical Model

- Defines the tools and techniques that will be required to follow the methodology framework.
- It will consist of the use of a combination of the information systems modelling views:
  - Data model,
  - Process model
  - Behavioural (or dynamic) model.
- CASE tools

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Managerial Model

- It provides the framework within which the development will take place.
- It defines stages, steps, tasks to be undertaken.
  - **Which** steps considered **essential** / which are optional
  - **When** they will be undertaken
    - the **order** in which the steps should be performed
  - **How** they should be managed
    - **How** developers and teams should work together
  - **What** will the the **end-deliverables be**

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Where we've got to today..

- Development techniques have changed rapidly over the years
- The approach to systems development continues to evolve
- As we utilise new technology and s/w development approaches

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There is some agreement

All software developers agree that the key is not the specific methodology but having a framework to structure your project effectively.

Today's developers have a much more realistic view: *you need to adapt your recipe to the ingredients you have and to the preferences of the consumers*

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Significant Developments in Methods and Frameworks over the years

- SSADM
- RUP
- UML (not a framework/method!)
- Agile
- XP, Scrum, DSDM

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Systems Development Life Cycle

The late 1960's saw the development of several *methods* and *frameworks* designed to overcome the problems associated with systems development projects.

Outlined the key activities forming the SDLC.

Modern development frameworks built from this.

Main criticisms: rigid, no back-tracking or iterations allowed

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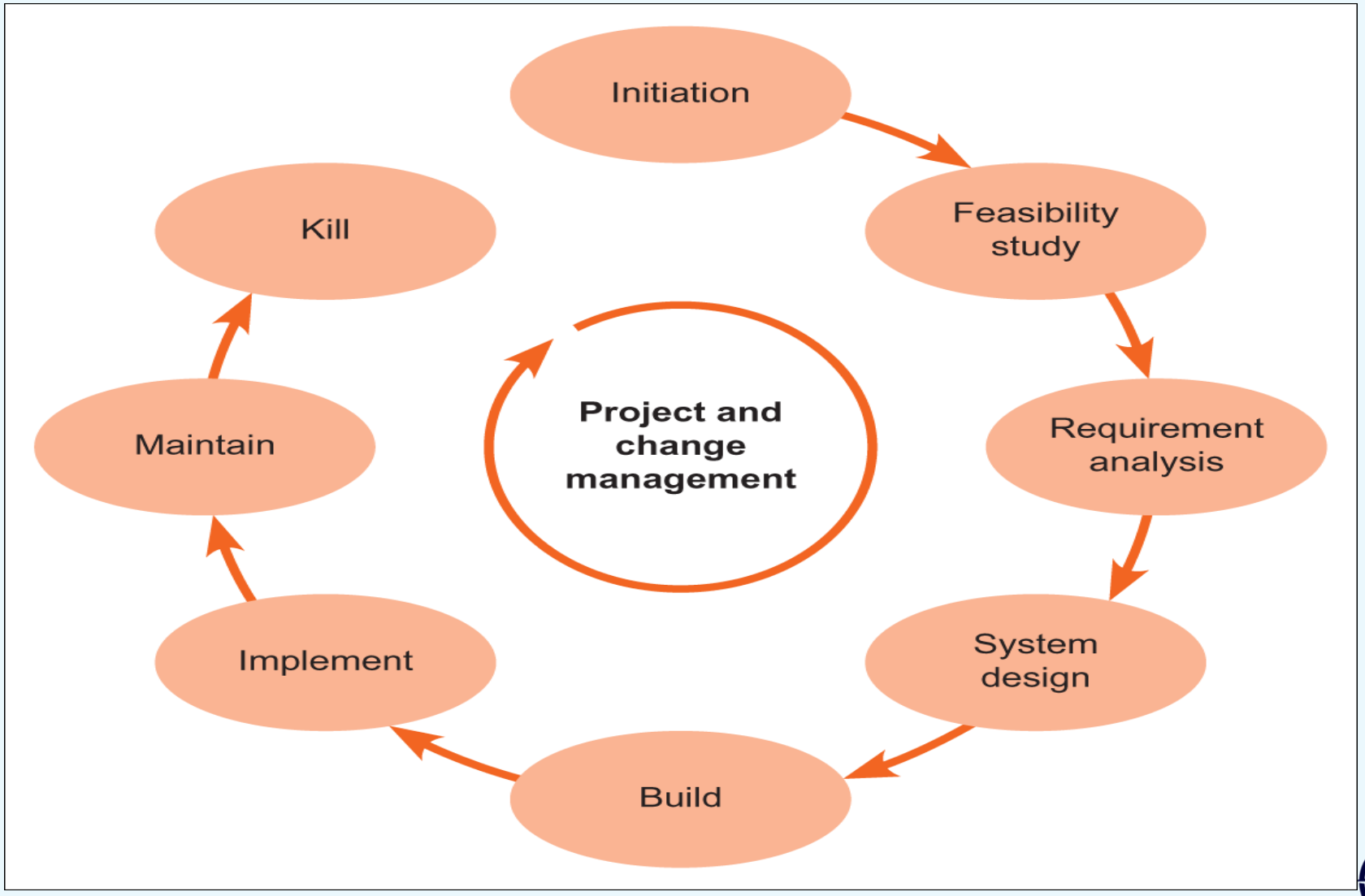
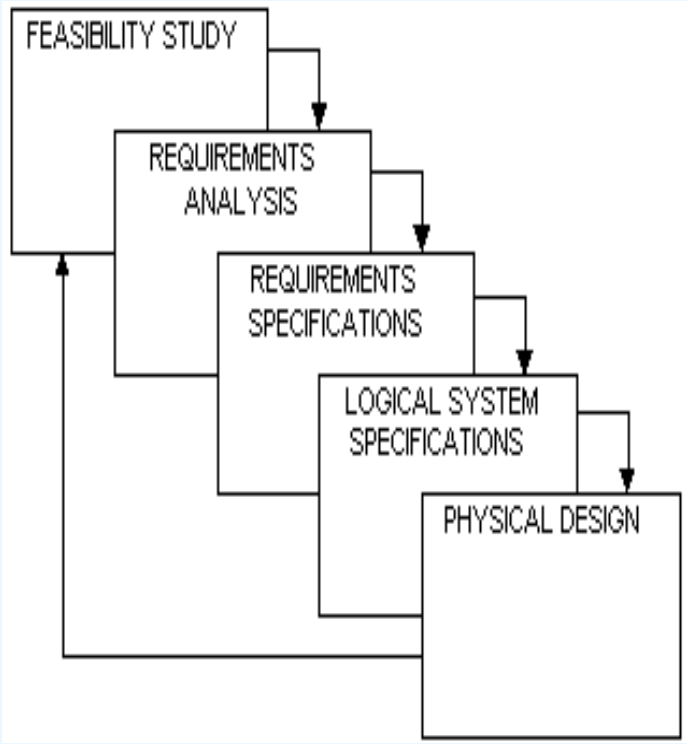
Systems Development Life Cycle



Structured Systems Analysis & Design Method (SSADM) Stages



Each stage has several steps

Each step may have several sub steps

Each stage/step/sub step has one or more end deliverables associated with it…..

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Example: SSADM *Stage 0 and 1*

Feasibility

- SLEPT factors
- Any preliminary models that have been constructed
- Details of rejected options and the reasons for their rejection.
- The product of this stage is a formal feasibility study document.

Investigation of current environment

- Establish analysis framework
- Investigate current processing
- Investigate current data
- Derive logical view of current services
- Combination of interviewing employees, circulating questionnaires, observations and existing documentation, the analyst comes to full understanding of the system as it is at the start of the project.
- Assemble investigation results

Example: SSADM - *Stage 2 and 3*

Business System options

- Brainstorming session
- Cost benefit analysis/ Impact of new “To Be” system
- Define and select business system options
- Define Requirements

Requirements Specification

- Logical specification of what the system will do (not how)
- Use of Logical models such as DFD, and LDM
- The product of this stage is a formal requirements specification document.

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Example: SSADM - *Stage 4 and 5*

Technical System options

- the hardware architectures
- the software to use
- the cost of the implementation
- the staffing required
- the physical limitations such as a space occupied by the system
- the distribution including any networks which that may require
- the overall format of the human computer interface
- The output of this stage is a chosen technical system option.

Logical Design

- the overall format of the human computer interface
- The product of this stage is the logical design which is made up of: Data catalogue, Required logical data structure, Logical process model

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Example: SSADM
Stage 6

Physical Design

- Logical specifications of the system are converted to descriptions of the system in terms of real hardware, software and to the appropriate standards.

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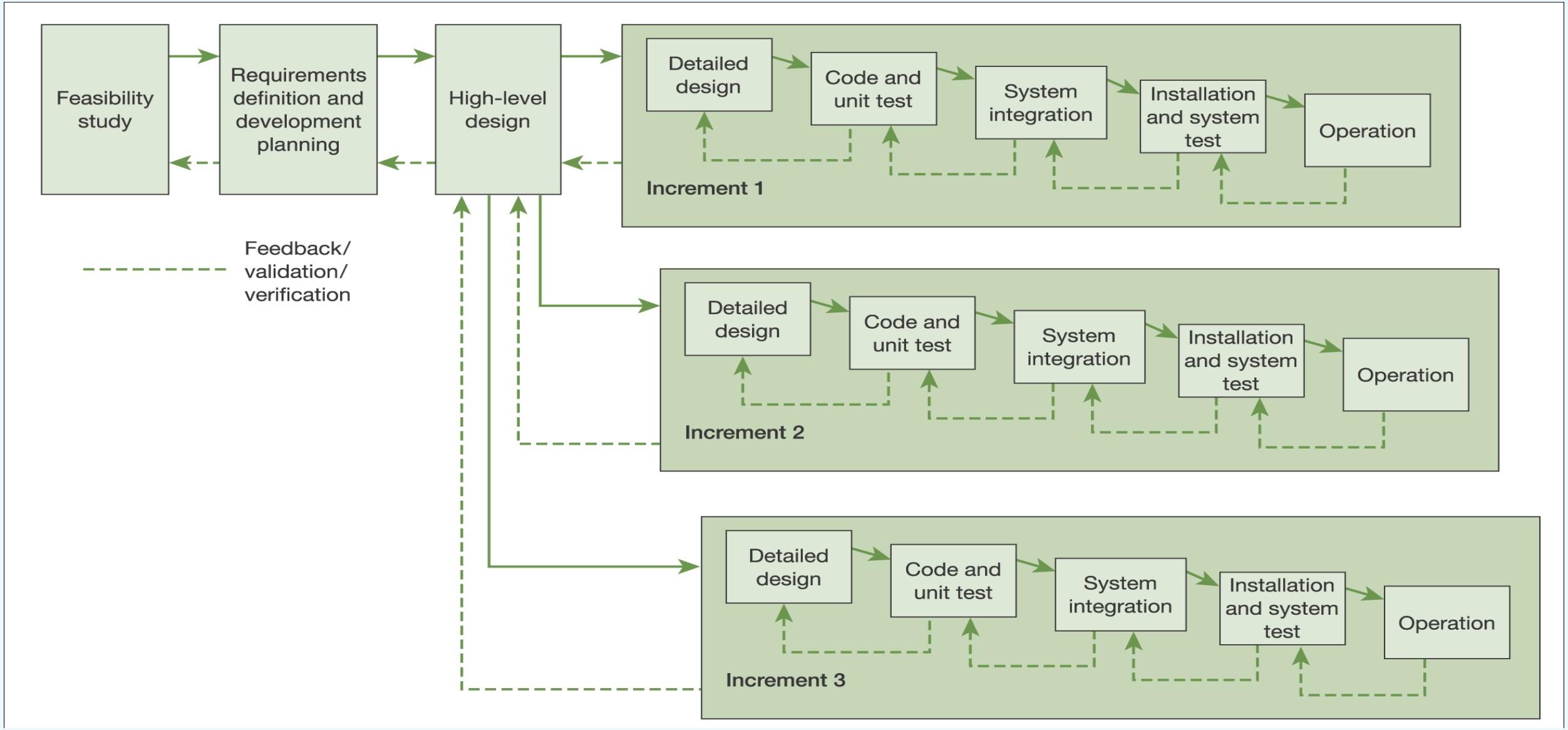

Incremental Model (early 1980s)

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Methodologies'

1. **RAD (Rapid Application Development) (1980s):**
    1. Focuses on rapid prototyping and quick feedback from end-users.
    2. Iterative development with user involvement throughout. (More in the next weeks)
2. **Object-Oriented Development (1980s-1990s):**
    1. Emphasizes encapsulation, inheritance, and polymorphism.
    2. Promotes modular design and code reuse.

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RUP - The *Rational Unified Process*

"A full-fledged process able to support the entire software development life-cycle".

"Use case driven, architecture centric, iterative and incremental"

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Use case driven

Capture *the user requirements*

Use case describes an *element of the functionality of the system*

All the use cases together (the use case model) depict the total functionality of the system

- The system is designed based upon the use cases
- Implemented to support the use cases
- Tested upon content of the use cases

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Architecture centric

End deliverables or ‘artefacts’ are very important in the RUP just

Compares s/w architecture to architecture of a house.

Many methods ignore development of the architecture (models etc) and concentrate on development of software.

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Iterative and Incremental

- Belief that user requirements cannot be fully and accurately defined at one go.
- Requirements evolve and change over time as understanding deepens.

- *This is a key concept of modern methods, and we will come back to it over and over again.*

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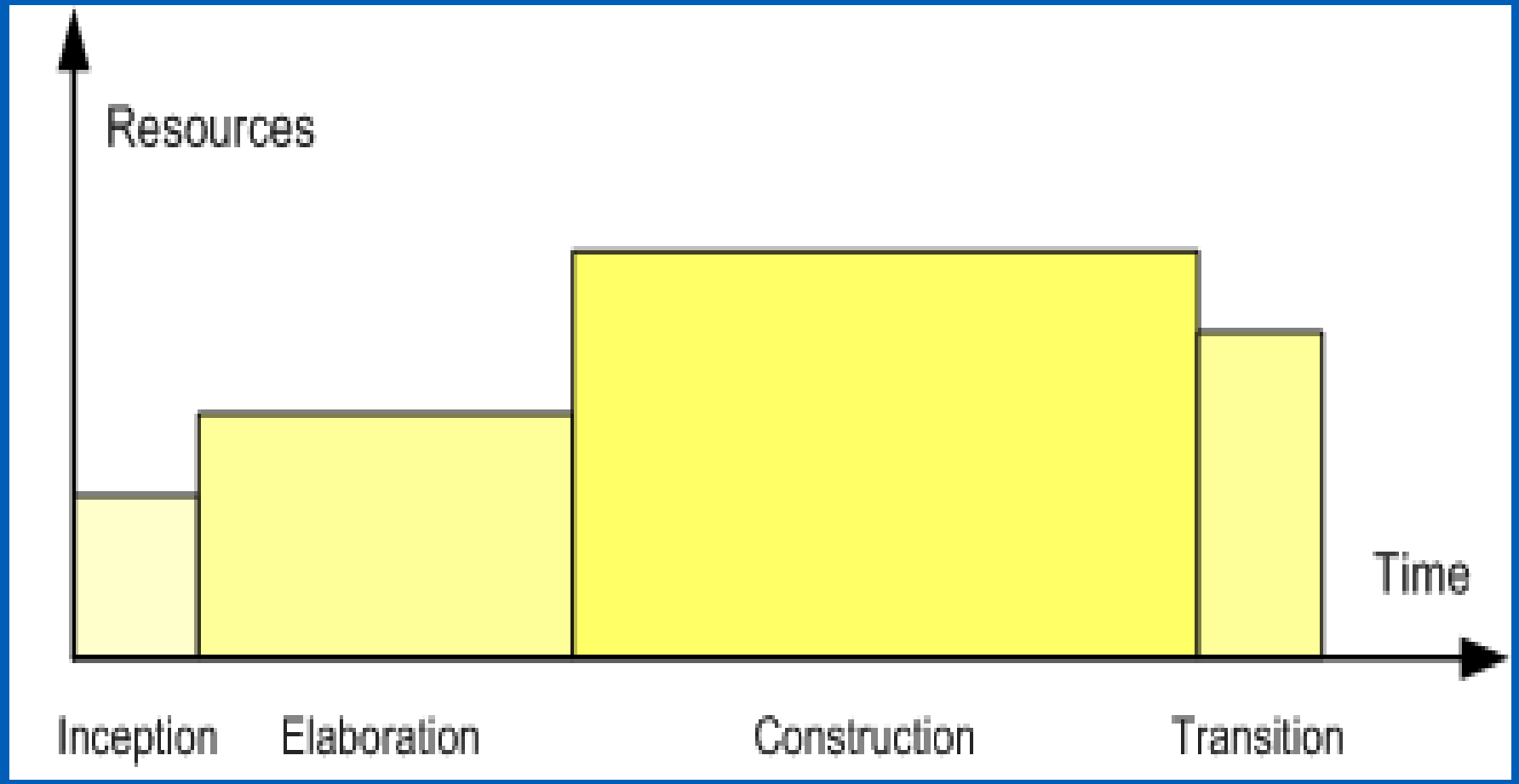
RUP – Phases and resources



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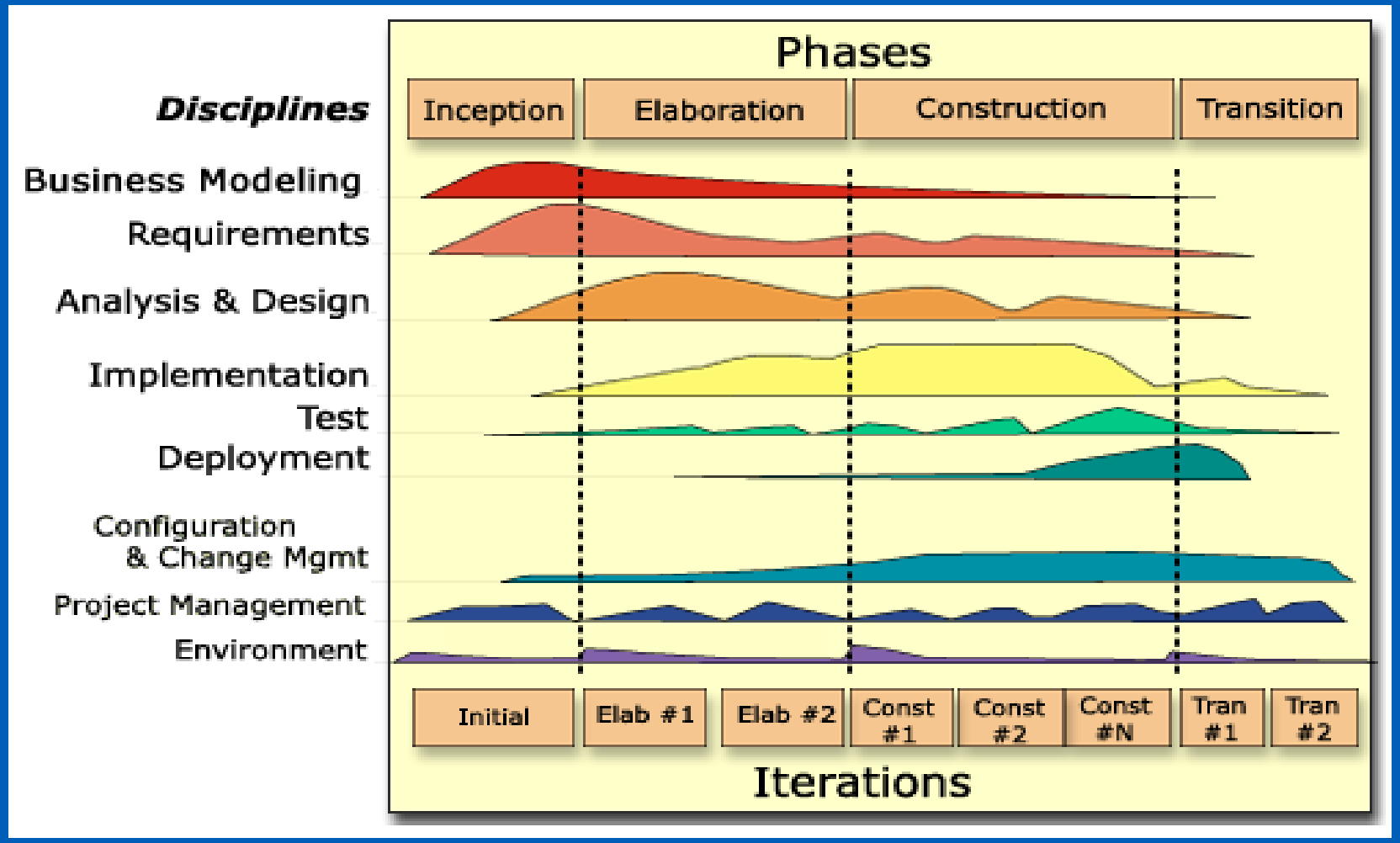
RUP – process structure



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Is it worth bothering with?

- Yes, it is standard for OO modelling
- Good thing is that certain models (e.g. use cases) can be used outside of OO modelling world
- Big criticism, though, is that it is too complex
- www.itjobswatch.co.uk rank it at 500
- Average salary around £65,000

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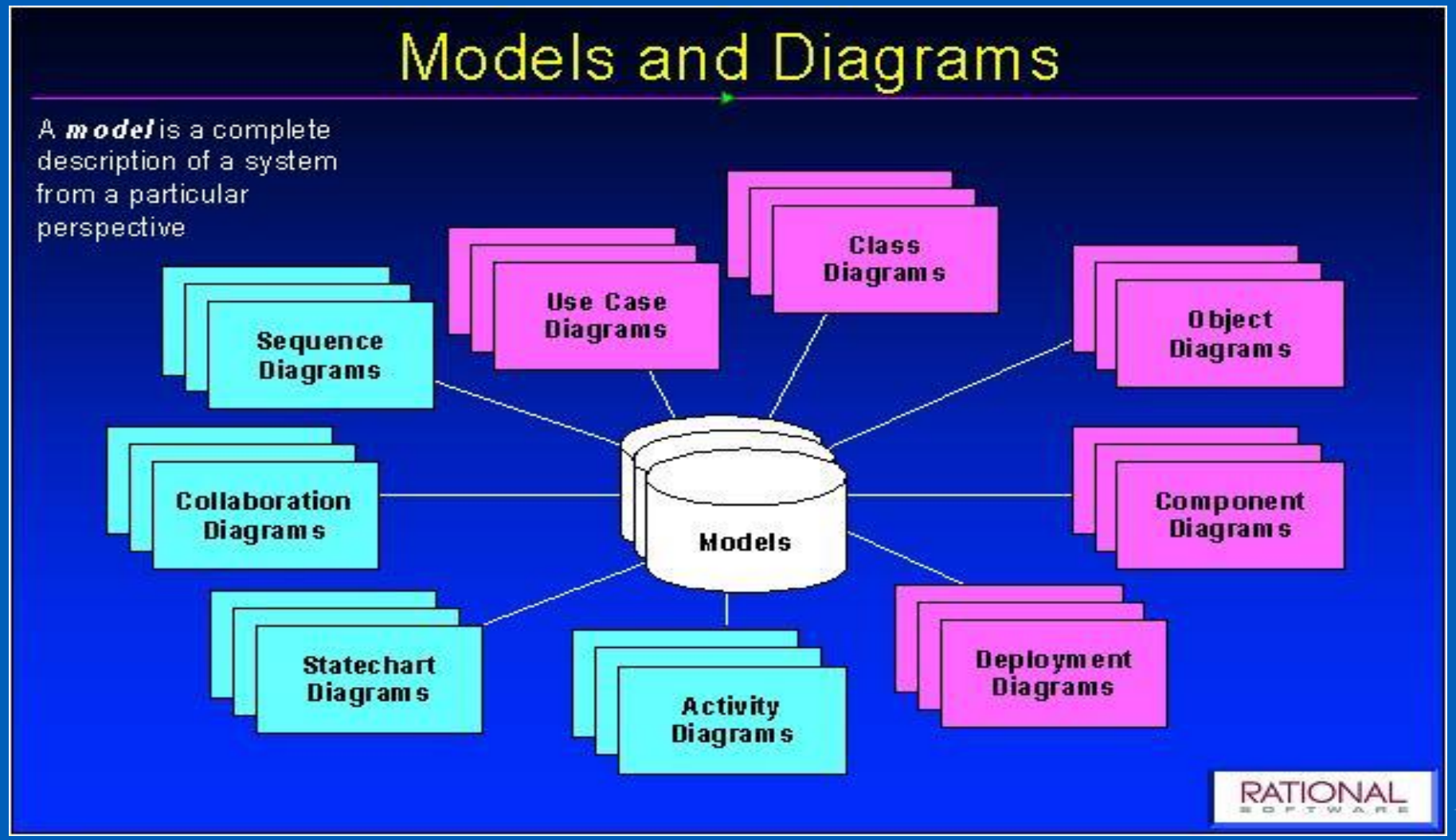
UML

- Unified Modelling Language
- Again, NOT a methodology based upon our definition
  - A set of specifications and design notations for object-oriented development
  - Developed combining modelling techniques developed by Booch, Rumbaugh and Jakobsen
- Lots of methods have been developed from it.

UML models and diagrams



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Is it worth bothering with?

- Yes, to an extent
  - Still used on large scale products
  - Owned by IBM
  - www.Itjobswatch.co.uk rank it at 422 in Jan 2024 (up from 982 in 2022)
  - Average salary around £65,000 (up from £60,000 last year)

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Agile Methods (1)

- Agile Framework is a family of methodologies, not a single approach to software development.
- They are based on the original concepts of Rapid Application Development (RAD)– more in coming weeks
- The idea has been developed to incorporate later stages of the life cycle (software development) and project management
- DSDM is an Agile Method developed in the UK

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Agile Methods (2)

- In 2001, 17 prominent figures in the field of agile development (then called "light-weight methodologies") came together to discuss the unifying theme of their methodologies.
  - Formed the Agile Alliance
  - Created the Agile Manifesto
- The background of these ‘experts’ was in software development rather than in requirements definitions/business analysis and so a new way of thinking for this discipline

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Agile Methods (3)

The Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

http://agilemanifesto.org/

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Agile Methods (4)

- Some of the well-known Agile software development methodologies include:
  - Scrum
  - DSDM
  - Feature Driven Development
  - Extreme Programming (XP)
  - Lean software development

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Key Agile concepts

- Most agile methods attempt to minimize risk by developing software in short timeboxes, called iterations.
- which typically last 1 to 4 weeks.
- Each iteration is like a miniature software project of its own and includes all the tasks necessary to release the mini-increment of new functionality: planning, requirements.

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Agile Methods (6)

- Agile methods emphasize face-to-face communication over written documents.
- Most agile teams are located in a ‘bullpen’ and include all the people necessary to finish software.
  - At a minimum, this includes programmers and their "customers."
  - The bullpen may also include testers, interaction designers, technical writers, and managers.

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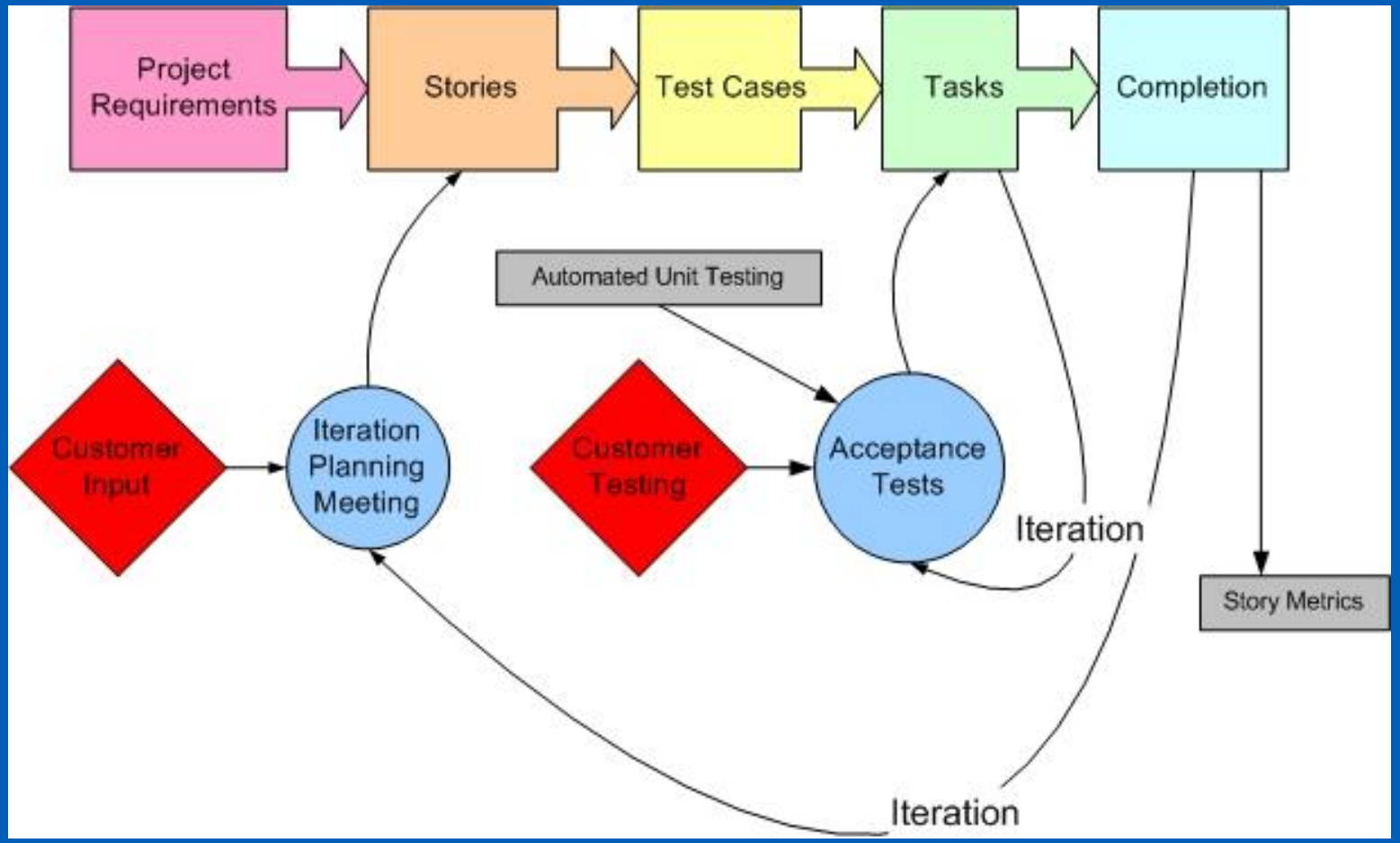
Extreme Programming



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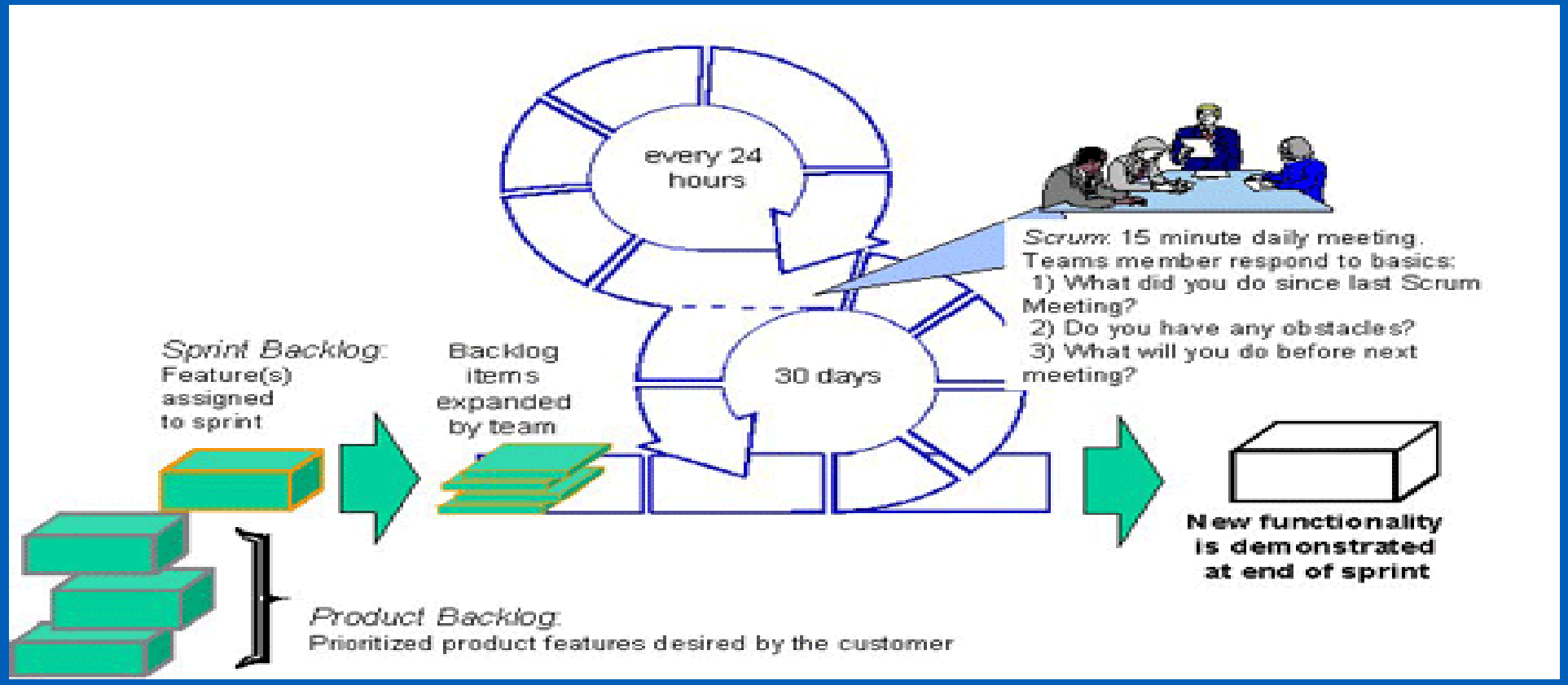
Scrum



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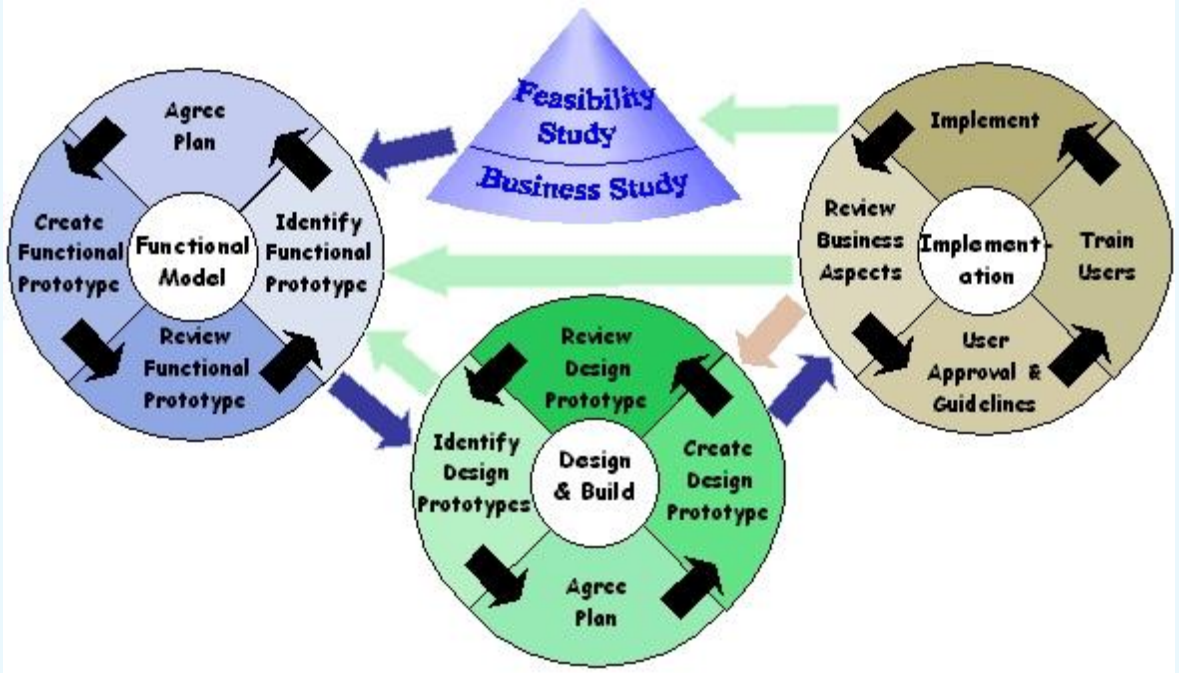
DSDM



- DSDM is the proven framework for Agile PM
- Created in 1994 by founding members of the DSDM consortium (Now Agile Business Consortium)
- 2001 DSDM consortium helped shape the Manifesto for Agile Software Development to create Agile Alliance.

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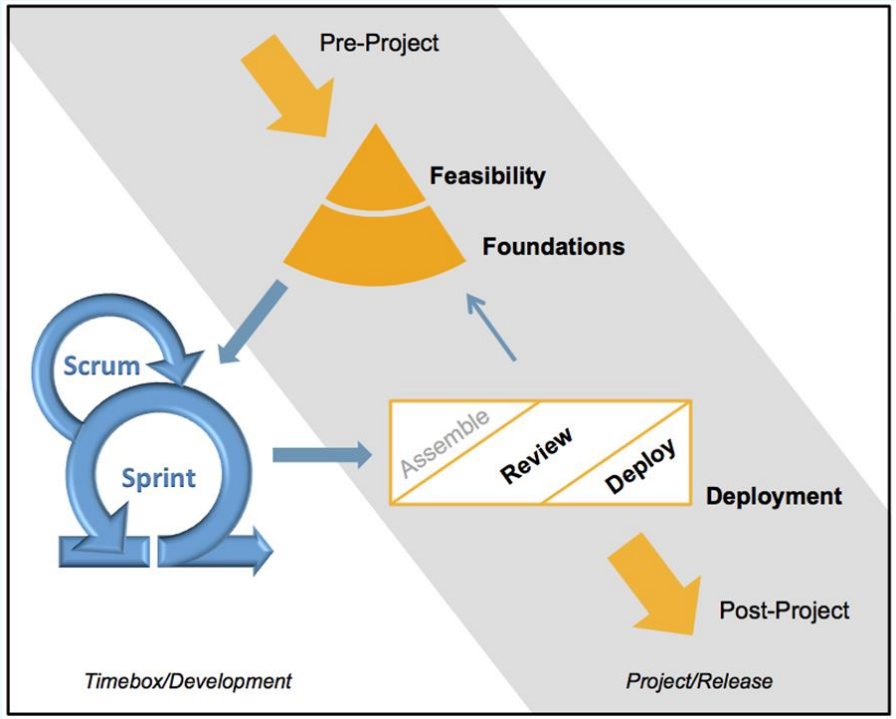
Agile PM with Scrum



- Created in 2014 Agile PM uses the DSDM process and the project lifecycle (https://www.agilebusiness.org)

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Agile development

Agile Software Development

- Ranked 2st Jan 2024
- Average salary around £65,000

(up from £62,500 the previous year)

Source:
https://www.itjobswatch.co.uk/default.aspx?q=&ql=Agile+Software+Development&l=&ll=&id=0&p=6&e=5&sortby=&orderby=

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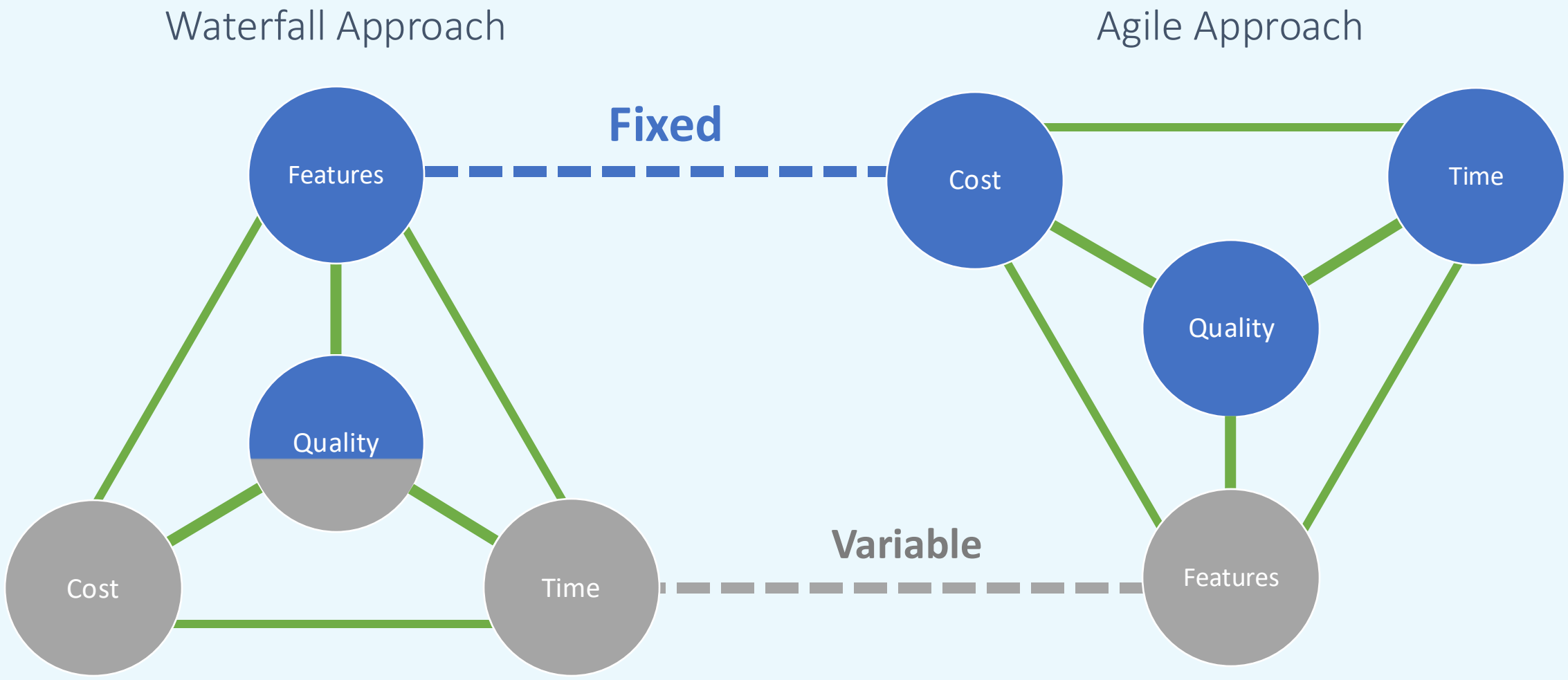
In a Nutshell



Potential Benefits Comparison

Agile Approach

- Faster development time with higher levels of productivity
- Flexibility to adapt to uncertain and changing requirements
- Faster learning curves to incorporate process improvements
- Addresses long-windedness of other approaches
- Prototyping used to:
  - assist users define requirements by demonstrating possibilities
  - using an approach where requirements and solutions evolve through collaboration
  - test performance implications

Waterfall

- Higher levels of control and predictability over cost and schedules
- Less dependent on highly skilled resources
- Consistent with many typical company environments – doesn't require significant change
- Higher levels of efficiency for products with low levels of uncertainty
- Works well when product is well-understood

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Potential Trade-offs

Agile	Waterfall

• Can be difficult to implement and might require retraining of the company staff
• Might require change management to changes company's culture if not consistent with an agile strategy
• May not be suitable if the company is large and complex
• Might not provide sufficient level of risk management and control for high-risk projects or regulated environments
	• May have overly bureaucratic control and documentation requirements that:
	  • Add unnecessary overhead, and
	  • Raise product development costs and result in excessively long product development times
	• Emphasis on control may impact the ability to react to emergent and changing business requirements
	• Inadequate emphasis on learning and continuous improvement

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- **Systems Development Life Cycle – Introduced 1960/70s**
  Focus: Linear, rigid, no backtracking
  Structured Systems Analysis and Design Methodology (SSADM) - **1980s**.
  Focus: Formalized, document-driven approach with data flow diagrams and system modelling.
- **Object-Oriented (OO) Methodologies** - Gained prominence in the **1990s**.
  Focus: Designing systems around objects (encapsulation, inheritance, and polymorphism).
- **Rapid Application Development (RAD)** - Developed in the **late 1980s to early 1990s**.
  Focus: Speed and flexibility, emphasizing iterative prototyping and user feedback.
- **Rational Unified Process (RUP)** - Introduced in the **1990s**.
  Focus: Iterative, risk-driven development tailored to specific project needs.

Dynamic Systems Development Method (DSDM) - Introduced in the **mid-1990s**.
Focus: Iterative and incremental approach, with principles emphasizing active user involvement and quality.

Agile - Manifested with the Agile Manifesto in **2001**.
Focus: Collaborative, adaptive, and iterative approaches, prioritizing working software and customer satisfaction.

Disciplined Agile (DA) - Introduced in the **2010s**.
Focus: A hybrid approach that combines Agile, Lean, and traditional methodologies to provide a flexible delivery framework.

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Short Quiz

- Go to **www.menti.com** and use the code **7862 4737**
- **https://www.menti.com/8f6vxnfv43**

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Summary

- Computer-related software projects may use Systems Development Life Cycle (SDLC), Iterative, Scrum, Kanban, DSDM, XP.
- SDLC was followed for large complex software projects.
- Methods such as RUP designed for larger scale systems are still used but they are much more flexible
- Both XP and Scrum are very customer driven, boast of frequent product or software releases and constant testing.

Summary (2)

- Agile (RAD) methods are currently in vogue – still criticism of them
- Agile methods like XP and Scrum have become popular with smaller teams for small projects.
- Current approaches advocate
  - Time boxes, working in small groups, integrating analysts, designers, programmers and users
- Agile Methods are driven towards collaborative teamwork and adapt well to organizational culture

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Thank you and questions?

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